Attention Heads are Complementary Visual Units: Mitigating Hallucinations in LVLMs via Adaptive Visual Cues Focusing

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Abstract

Recent studies have explored attention dynamics in Large Vision-Language Models (LVLMs). However, most existing approaches aggregate attention across heads, potentially obscuring head-specific visual information and weakening visual grounding, leading to hallucinations. In this work, we revisit hallucination from the perspective of how visual information is distributed and utilized across attention heads. We observe that a small subset of visual tokens accounts for most of the attention within each attention head, with these tokens—defined as head-wise visual cues—being complementary across heads, suggesting that effective grounding requires preserving head-specific support. We further find that the proportion of visual cues within visual attention declines during generation, leading to key visual information loss and hallucination. In addition, hallucinated tokens show weaker utilization of long-range textual context compared to correctly generated tokens. Building on these findings, we propose **V**isual Cu**e**s **R**einforcement via Head-Adaptive **D**e-redundancy and Distance-Aware **A**ttenuation **(VerDA)**, a training-free method to mitigate hallucinations. Specifically, we remove redundant visual tokens via head-adaptive de-redundancy, suppress local textual bias through distance-aware attenuation, and reinforce visual cues by redistributing attention. Experiments across multiple LVLMs and benchmarks demonstrate that VerDA consistently outperforms existing methods in mitigating hallucinations, with negligible inference overhead. The code is available at https://github.com/huazhenglin2003/VerDA.

1 Introduction

Large Vision-Language Models (LVLMs) [1, 2, 3, 4, 5] achieve significant progress in various vision-language tasks such as image captioning [6] and visual question answering [7]. However, LVLMs still face the challenge of hallucination [8, 9, 10], where the generated text is inconsistent with the visual input. This problem significantly limits their deployment in real-world applications [11, 12].

To mitigate hallucinations, several approaches incorporate external knowledge bases for alignment training [13, 14] or fine-tuning [15]. However, these methods often incur high computational and storage costs. In contrast, recent studies have focused on training-free approaches during the inference

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